



Screen-Based vs. Non-Screen Leisure: How Social Context Shapes Momentary Happiness and Stress

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Introduction

Leisure activities make up an important part of everyday life, but not all leisure may relate to well-being in the same way. Screen-based leisure, such as watching television or using a computer for fun, is often treated as interchangeable with other free-time activities, even though it may differ from non-screen leisure like reading, and arts and crafts. At the same time, social context may matter just as much as activity type. Doing an activity alone may feel different from doing it with other people. This project asks whether momentary happiness and stress differ across these kinds of leisure episodes, and whether those differences depend on whether a person is alone or with others.

Research Question

How do momentary happiness and stress differ between screen-based and non-screen leisure activities, and do those patterns change depending on whether people are alone or with others?

Data

This project uses the American Time Use Survey (ATUS) Well-Being Module for 2010, 2012, and 2013. The ATUS collects information on how people in the United States spend their time, and the Well-Being Module records momentary emotional ratings for selected activity episodes. I combined ATUS activity, well-being, respondent, roster, and Who files to create an episode-level dataset. The final analytic sample contains 12,444 respondents and 14,986 leisure activity episodes. I excluded episodes with invalid happiness or stress values and kept only episodes classified as either screen-based or non-screen leisure.

Variables / Sample Description

Key Variables

The two outcomes are momentary happiness and momentary stress, both measured on a 0-6 scale. The main predictors are activity type (screen-based vs. non-screen leisure) and social context (alone vs. with others). Screen-based leisure includes television, leisure computer use, and games, while non-screen leisure includes reading, arts and crafts, and relaxing. I also included age, sex, and employment status as control variables in the regression models.

Sample Description

The final sample includes more screen-based than non-screen episodes overall. Among the four comparison groups, the largest group was screen-based leisure with others (8,138 episodes) and the smallest was non-screen leisure alone (1,262 episodes). The sample had a mean age of about 50.3 years, and the analysis also includes variation in employment status and sex.

Methods

- Merged ATUS activity, well-being, respondent, roster, and Who files into an episode-level dataset.
- Created 6-digit activity codes and grouped selected activities into screen-based and non-screen leisure.
- Defined social context as alone versus with others using the Who file.
- Calculated weighted mean happiness and stress for four groups: non-screen alone, non-screen with others, screen-based alone, and screen-based with others.
- Estimated WLS models with clustered standard errors and mixed-effects models with respondent random intercepts.
- Included controls for age, sex, and employment status.
- Compared WLS and mixed-effects results to assess robustness to within-respondent dependence

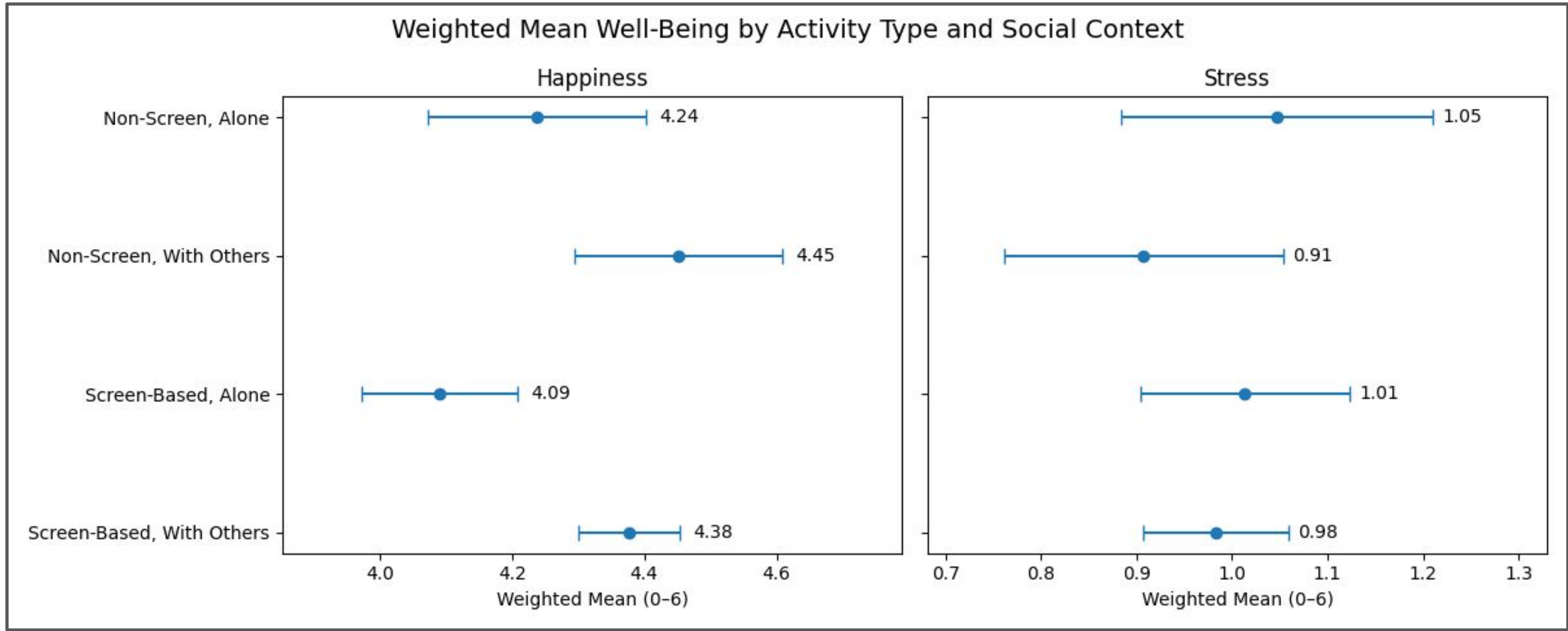


Figure 1. Group differences were clearer for happiness than for stress. Non-screen leisure with others had the highest happiness, while screen-based leisure alone had the lowest.

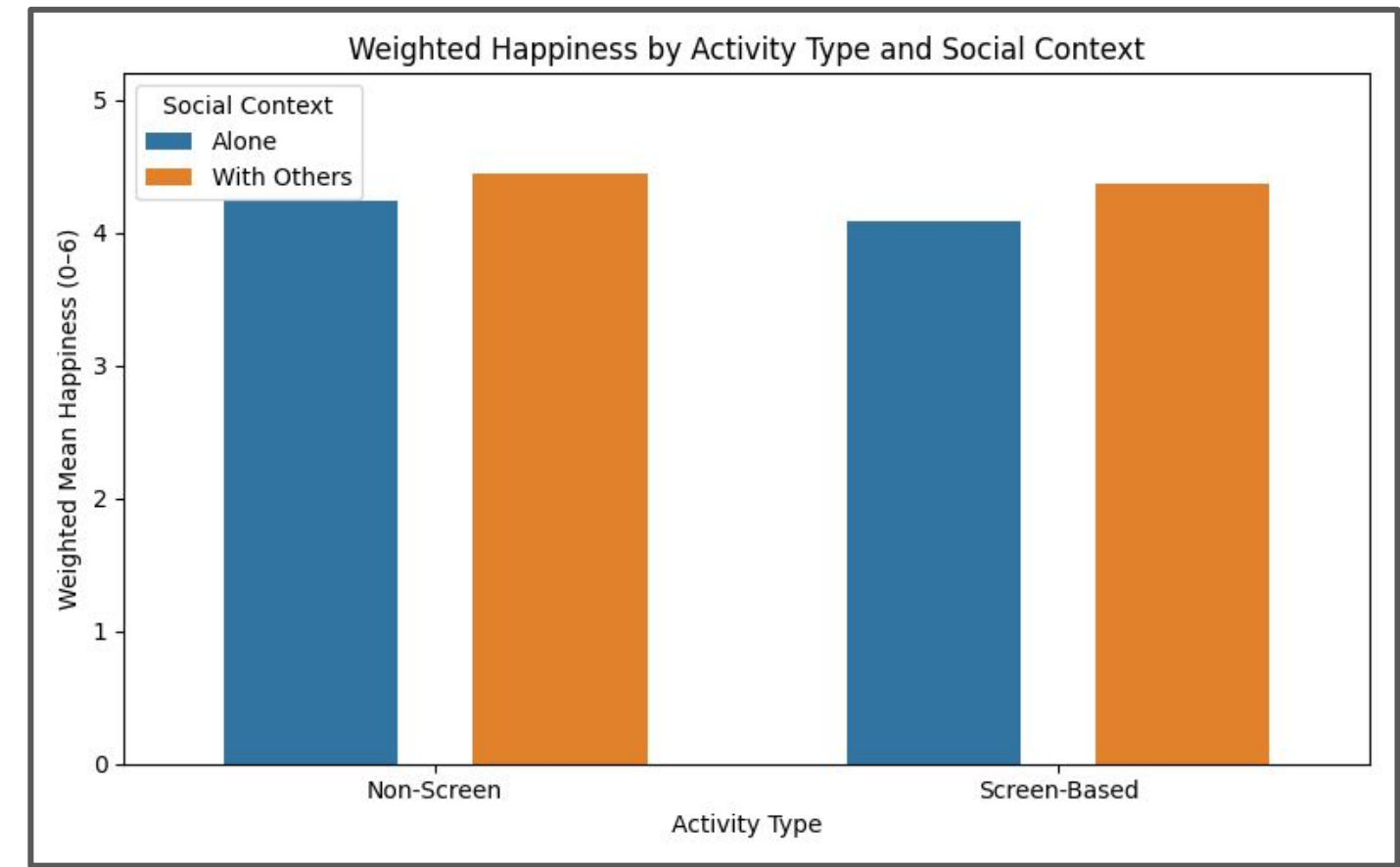


Figure 2. Weighted happiness was highest for non-screen leisure with others and lowest for screen-based leisure alone. Differences were clearer for happiness than for stress.

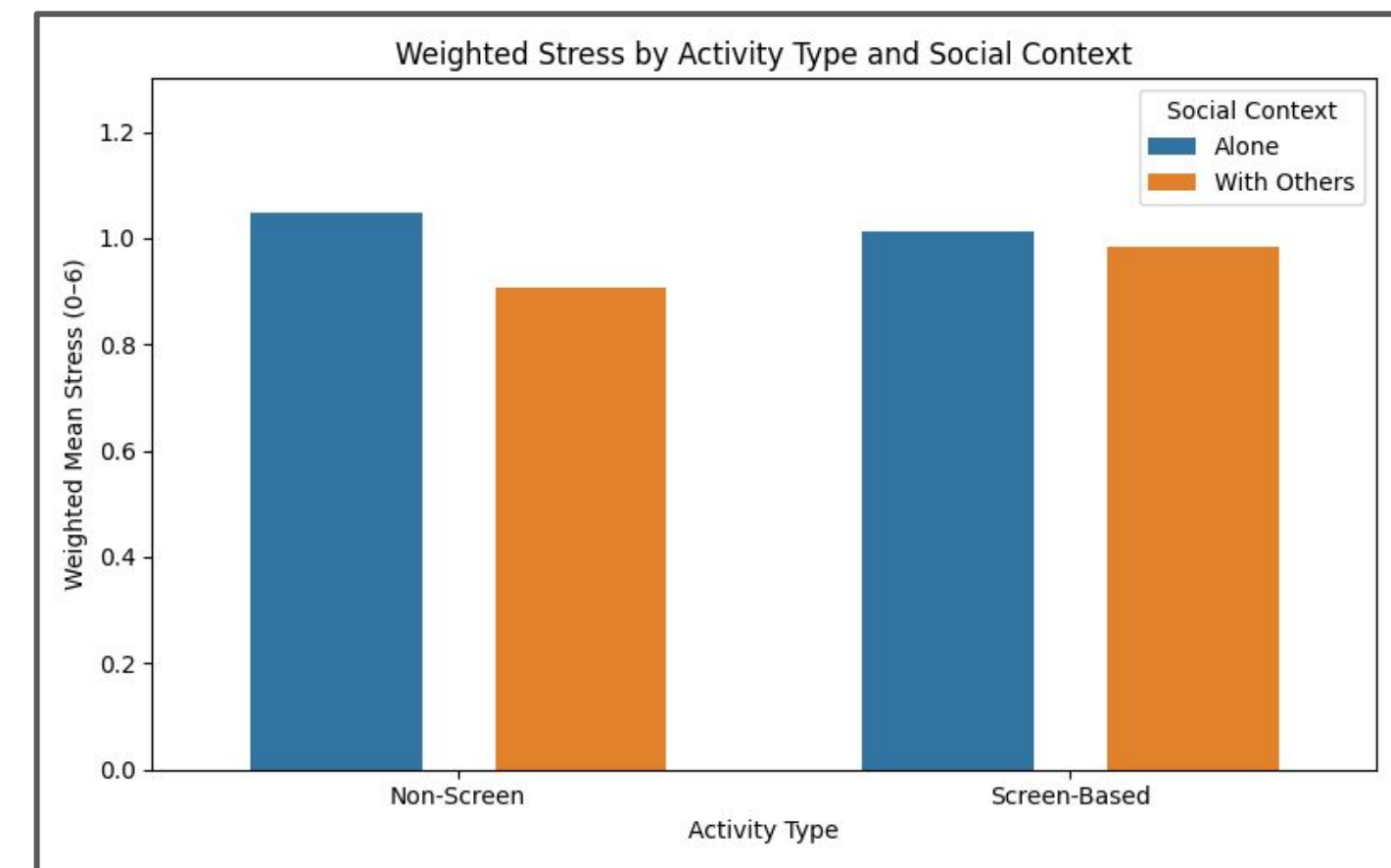


Figure 3. Stress differed less across groups than happiness. Non-screen leisure with others had the lowest weighted stress, while non-screen leisure alone had the highest.

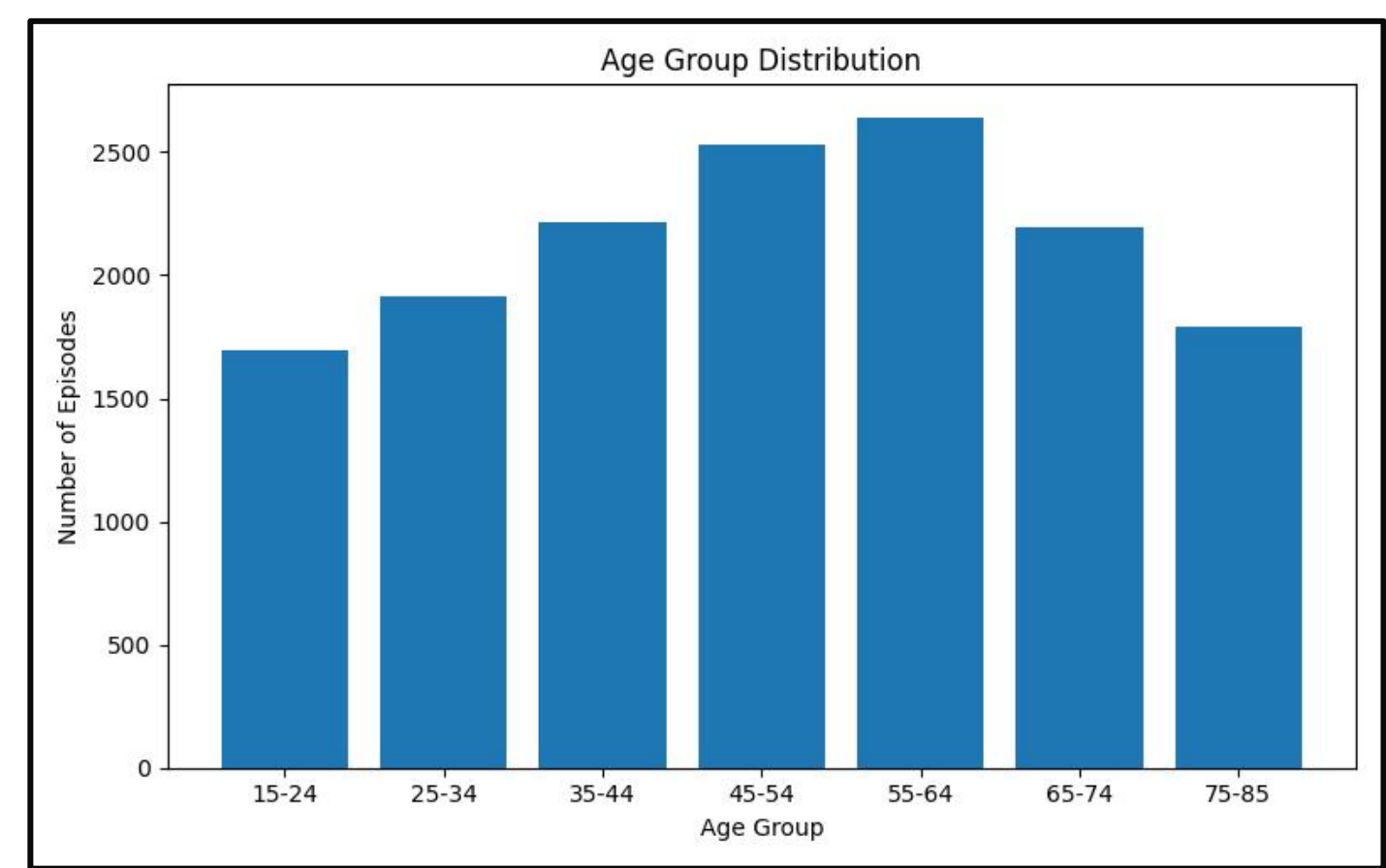


Figure 4. The analytic sample included episodes across all age groups, with the most episodes from ages 45–64.

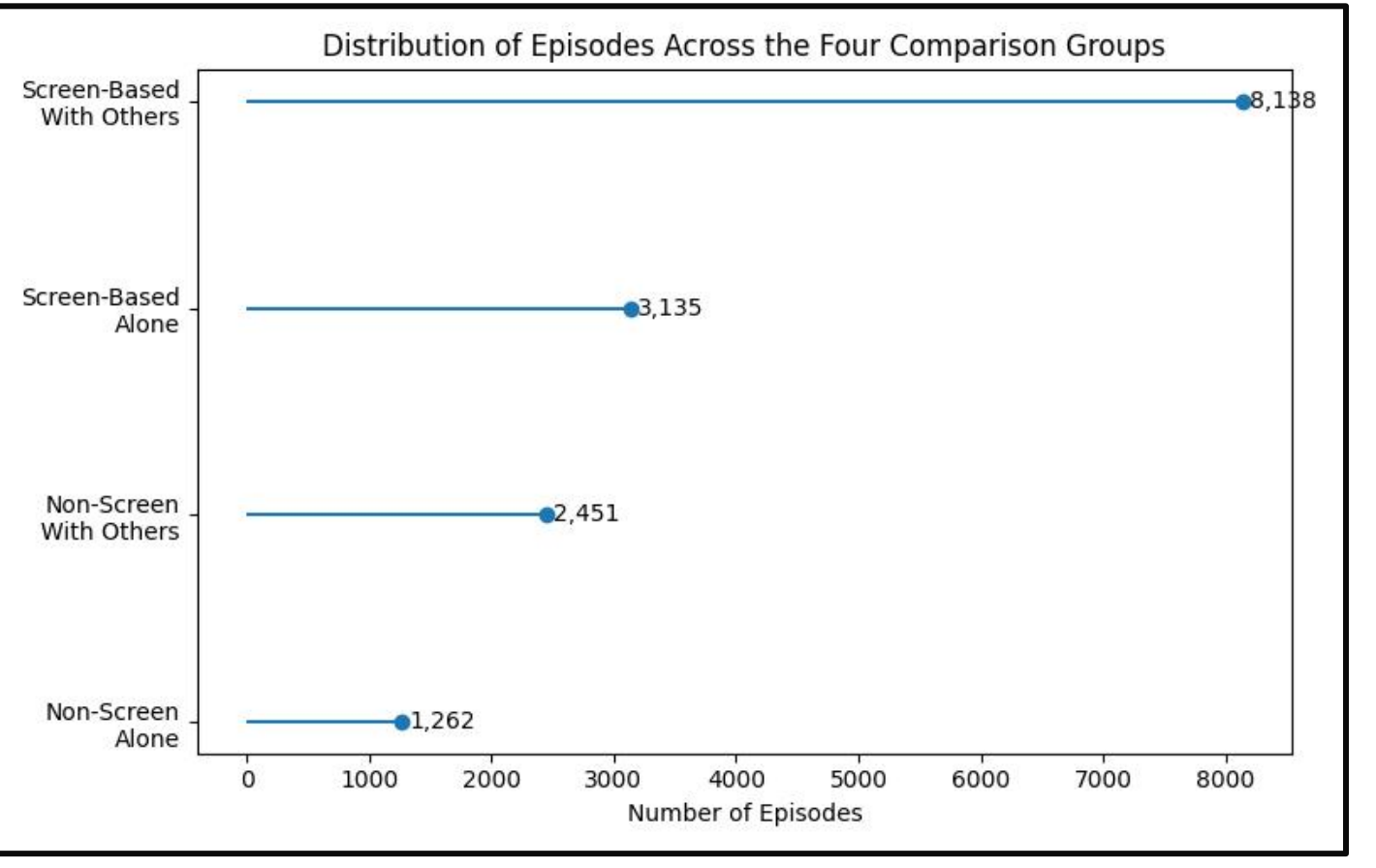


Figure 5. Screen-based leisure with others was the largest group in the sample, while non-screen leisure alone was the smallest.

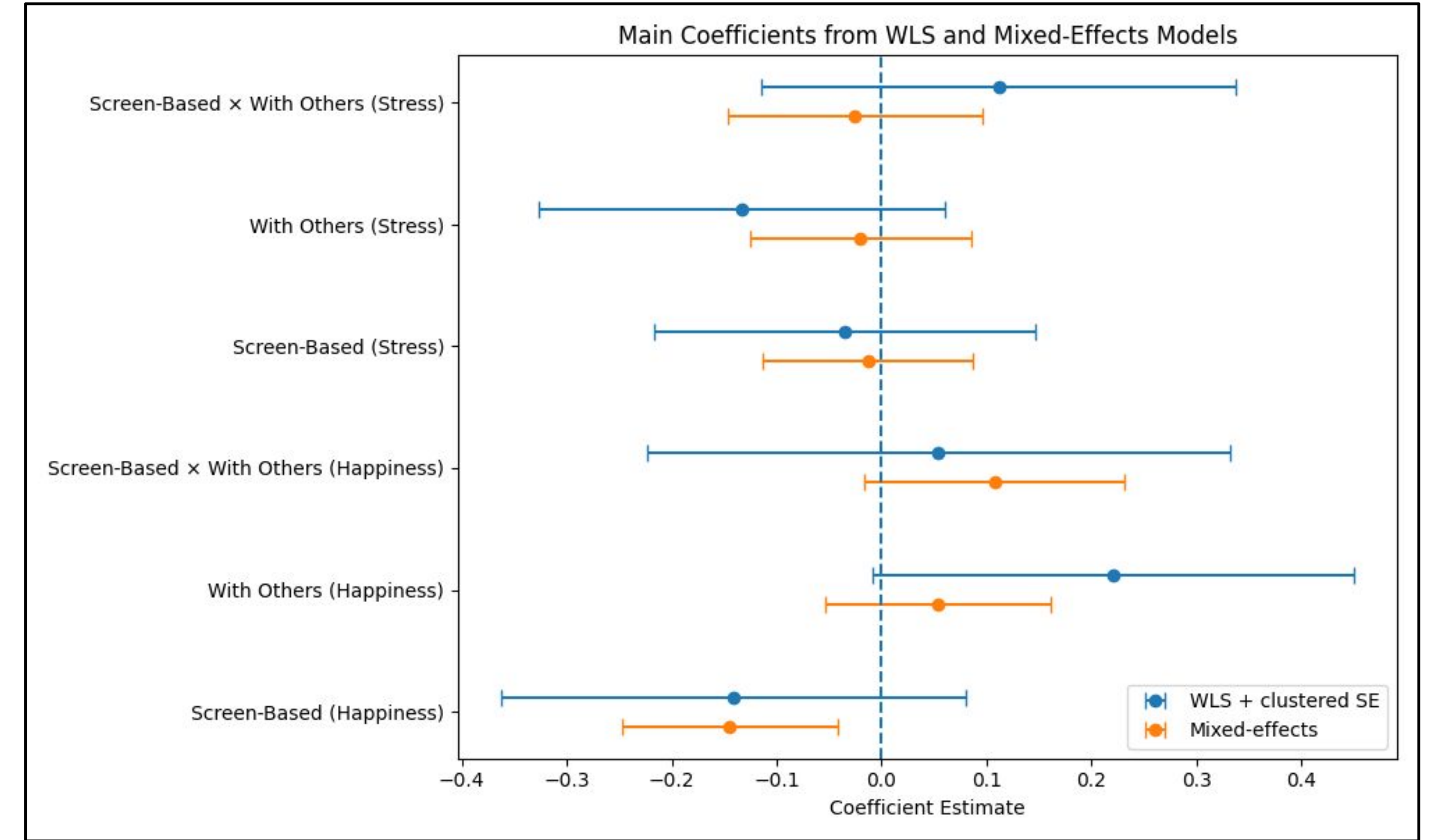


Figure 6. Coefficients were generally clearer for happiness than for stress, and the WLS and mixed-effects models showed similar overall patterns.

Results

Outcome	Term	WLS + clustered SE	Mixed-effects
Happiness	Screen-Based	-0.141 (p=.212)	-0.144** (p=.006)
	With Others	+0.221 (p = .060)	+0.054 (p = .327)
	Screen × Others	+0.054 (p=.701)	+0.108 (p=.087)
	Sig. controls	TELF5: -0.208** (p=.002)	TESEX2: +0.099** (p=.001); TELFS4: -0.136* (p=.034); TELFS5: -0.122*** (p<.001)
	Fit	R ² =.010; Adj. R ² =.009	Marg. R ² =.004; Cond. R ² =.575
Stress	Screen-Based	-0.035 (p=.708)	-0.013 (p=.800)
	With Others	-0.133 (p=.177)	-0.020 (p=.708)
	Screen × Others	+0.112 (p=.332)	-0.025 (p=.682)
	Sig. controls	TELF4: +0.347** (p=.006); TELFS5: +0.230*** (p<.001)	TESEX2: +0.080** (p=.007); TELFS4: +0.308*** (p<.001); TELFS5: +0.168*** (p<.001); Age: -0.002* (p=.019)
	Fit	R ² =.008; Adj. R ² =.007	Marg. R ² =.004; Cond. R ² =.636

Model Results

- Results were clearer for happiness than for stress.
- In the mixed-effects happiness model, screen-based leisure was associated with lower happiness than non-screen leisure (-0.144, p = .006).
- In the WLS happiness model, With Others was positive (+0.221) but only marginally significant (p = .060).
- The happiness interaction was positive in both models, which would mean that being with others may slightly lessen the happiness disadvantage of screen-based leisure, but this result was not statistically significant.
- For stress, the main predictors were weak and inconsistent across models, with no statistically clear effects for activity type, social context, or their interaction.
- Among the controls, employment status was the most consistent. Respondents in TELFS = 4 and TELFS = 5 tended to report lower happiness and higher stress, suggesting that employment instability may be linked to worse momentary well-being.
- Female respondents showed slightly higher predicted happiness and stress in the mixed-effects models, while older respondents reported slightly lower stress.

Model Fit

- WLS adjusted R² values were low, indicating that the predictors explained only a modest share of episode-level variation.
- In the mixed-effects models, marginal R² values were low, but conditional R² values were much higher. This means that fixed effects alone explained relatively little variation, but once respondent-level random intercepts were included, the models explained substantially more variation overall. The gap suggests that a large share of the variation reflects stable differences across respondents, not just differences across leisure episodes.
- R² and adjusted R² were very similar in the WLS models, suggesting little evidence of overfitting.

Conclusion

- Differences in leisure-related well-being appeared more clearly for happiness than for stress.
- Descriptively, non-screen leisure with others had the highest happiness, while screen-based leisure alone had the lowest.
- In the regression models, the clearest finding was that screen-based leisure was associated with lower happiness in the mixed-effects model.
- The stress results were small and inconsistent across models.
- Overall, leisure type and social context showed a clearer relationship with momentary happiness than with momentary stress.

Limitations

- This analysis is observational and should not be interpreted causally.
- Well-being measures were recorded only for selected activity episodes.
- Findings depend partly on how leisure activities were grouped and defined.
- Happiness and stress were self-reported at a single moment in time.
- Social context was simplified to alone vs. with others.

References

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